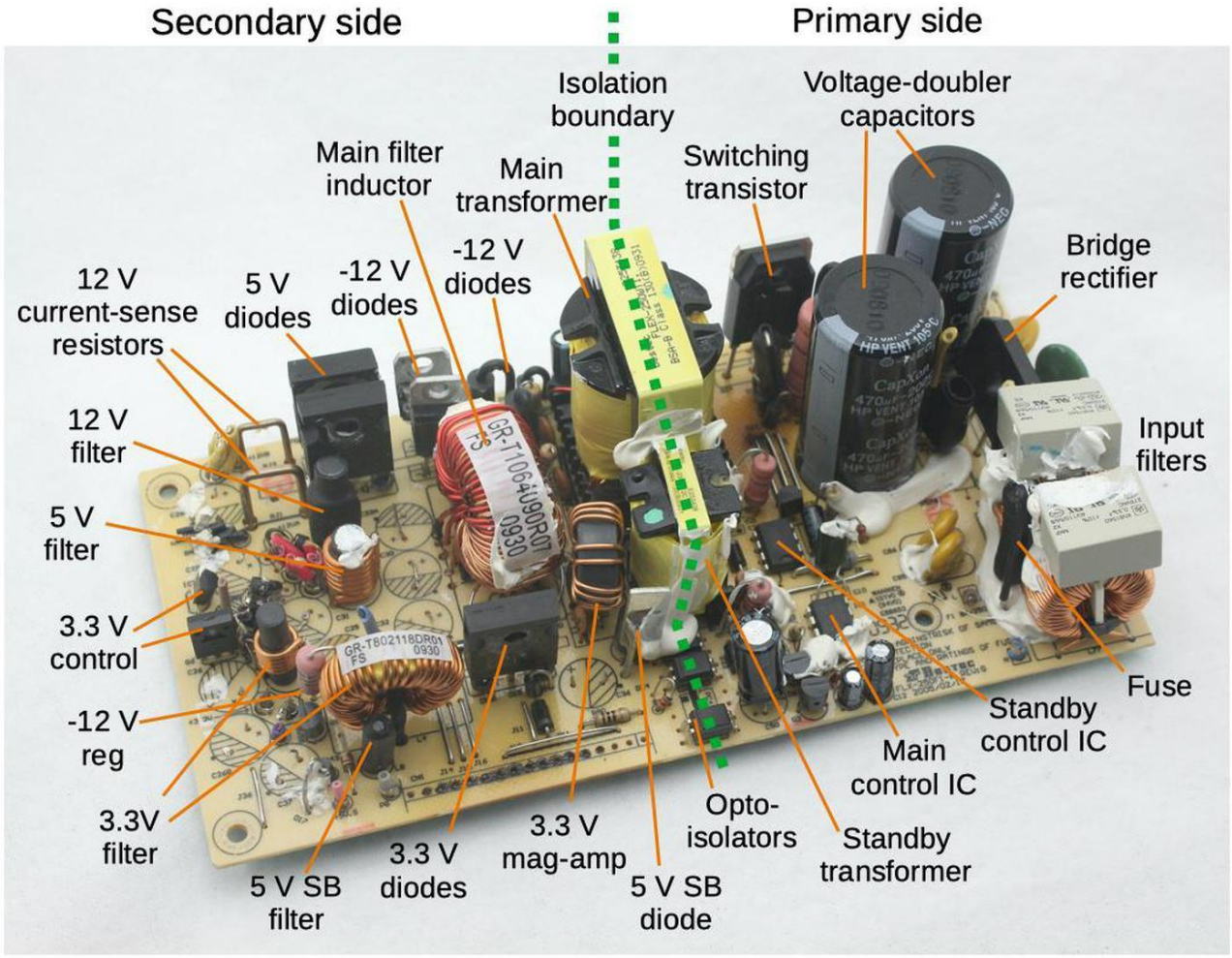
LAB 02

Subject

Identification of PC Power Supply (SMPS) Components



Key Parts Inside a PC Power Supply

1. **AC Input & EMI Filter**
    - Where the mains power enters the PSU.
    - The EMI filter (capacitors & coils) reduces electrical noise from the mains.

2. **Fuse**
    - Protects the PSU from sudden overcurrent or short circuits.
    - Always check fuse first if PSU won't power ON.
3. **Rectifier & Bridge Rectifier**
    - Converts AC to DC.
    - First stage after the EMI filter.
4. **Power Factor Correction (PFC) Components**
    - Large inductor and capacitors that optimize input current and improve efficiency.
5. **Switching Transistors / MOSFETs**
    - High-speed switches that chop DC to high frequency for transformer input.
6. **Main Transformer**
    - Steps down voltage and isolates output from mains input.
    - Heart of the SMPS.
7. **Output Rectifier & Filter**
    - Diodes and capacitors that convert transformer AC output back to smooth DC.
8. **Voltage Regulator / Controller IC**
    - Monitors and stabilizes output voltages; provides PWM control.
9. **Capacitors**
    - Store and smooth DC voltages on both primary and secondary sides.
10. **Cooling Fan**
    - Prevents overheating by circulating air inside PSU.
11. **Connectors & Output Wires**
    - Supply +12V (yellow), +5V (red), +3.3V (orange), ground (black) to motherboard and peripherals.

How These Parts Work Together

- **AC mains → EMI Filter → Rectifier + Filter → Switching Stage → Transformer → Output Rectifier + Filter → DC Outputs.**

This sequence shows the major stages converting wall AC into regulated DC rails that your PC components depend on.